# Acquisition of distributivity

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10-05-2023

- during language acquisition children can produce many errors
- but the textbook generalization (see Pinker (1995) a.o.): the fonological, morphological and syntactic acquisition ends (succesfully) before elementary school
- this talk: the semantic acquisition lasts much longer

- focus of the talk: acquisition of distributivity
- joint work with Žaneta Šulíková & Martin Juřen

**(a)** Žaneta

**(b)** Martin

**Figure 1:** Co-authors

- some linguistic terminology

(1) Two writers wrote three books.

a. distributive reading: ✍️$_A$ → 📗, 📙, 📘; ✍️$_B$ → 📕, 📓, 📒

b. cumulative reading: ✍️$_A$ → 📗, 📙; ✍️$_B$ → 📕

c. collective reading: ✍️$_A$ + ✍️$_B$ → 📗, 📙, 📘

Some expressions (singular universal quantifiers like *each*, *jedes*, *každý*, ...) are obligatorily interpreted distributively:

(2) Two writers wrote three books.

a. distributive reading: ✍$_A$ → 📗, 📙, 📘; ✍$_B$ → 📕, 📓, 📒

b. #cumulative reading: ✍$_A$ → 📗, 📙; ✍$_B$ → 📕

c. #collective reading: ✍$_A$ + ✍$_B$ → 📗, 📙, 📘

- in adult grammar

- but children acquire the right distributive interpretation late: Geurts (2003); Musolino (2009); Pagliarini, Fiorin, and Dotlačil (n.d.); Koster, Spenader, and Hendriks (2018) a.o.
- this was observed for Italian, English, Dutch (at least)

Illustrative example from Koster, Spenader, and Hendriks (2018)

(3) Elke jongen wast een boot.
Each boy.SG wash.SG a boat.SG
'Each boy is washing a boat.'

| | | | | |
|---|---|---|---|---|
| Elke | jongen | wast | een | boot. |
| Each | boy.SG | wash.SG | a | boat.SG |

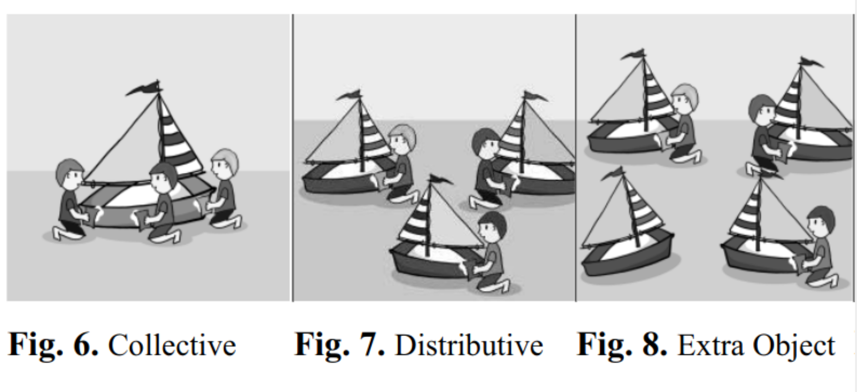



**Figure 2:** Example image conditions from Koster, Spenader, and Hendriks (2018)

Errors made by children:

- in Dutch (and all other observed languages)
- until age 8 (and sometimes after)

1. children (correctly) accept *elke* with distributive interpretation
2. children (incorrectly) accept *elke* with collective interpretation
3. children (incorrectly) reject *elke* with scenes including extra object (boat)

- non-trivial error: it is general lack of understanding of distributivity, collectivity and cumulativity (not only with universal quantifiers)
- universal quantifier is one of the functional words: essential parts of the human grammar machinery

# The acquisition errors I: spreading errors

- recall: children (≤ 9) incorrectly reject universal quantifier sentences with extra objects
- well-documented (Crain et al. 1996; Drozd 2001; Roeper, Strauss, and Pearson 2006)
- theoretical explanation (Geurts 2003): children interpret strong quantifiers as weak ones

## Strong vs. weak quantifiers

- distinction already from Barwise and Cooper (1981)
- linguistic test: appearance in existential constructions:

(4) a. There are {some/three boys } in the garden.
    b. There is {#each/every boy } in the garden.

- logical difference: weak quantifiers are symmetrical and strong quantifiers are relational

(5) a. Some boys dance.

(i) $\exists x[\mathbf{boy}(x) \wedge \mathbf{dance}(x)] = \exists x[\mathbf{dance}(x) \wedge \mathbf{boy}(x)]$

b. Each boy dances.

(i) $\forall x[\mathbf{boy} \rightarrow \mathbf{dance}(x)] \neq \forall x[\mathbf{dance}(x) \rightarrow \mathbf{boy}(x)]$

- Geurts (2003): tripartite structure of strong quantifiers is more cognitively demanding
  - children interpret all quantifiers as weak until they acquire the adult stage

- Geurts (2003): children make spreading errors since they interpret the restrictor by pragmatics

(6) a. Each [$_{\text{restrictor}}$ boy] [$_{\text{nuclear scope}}$ dances].
b. Each [$_{\text{restrictor}}$ boy] is washing a boat. correct
c. Each [$_{\text{restrictor}}$ boat] is washed by a boy. spreading error

Musolino (2009): 4-6 years old English speaking children

- truth value judgment task (sentence + picture) - children incorrectly interpret *each*:

(7) Three boys are holding each balloon.

- against the picture on the next slide (90% acceptance vs. 31% acceptance in adults control group)

**Figure 3:** False distributive interpretation

- Musolino's explanation: wrong identification of the restrictor like in (8)

(8) Each of the three boys is holding a balloon.

## Problems of Musolino's account

- the mechanism by which children identify the restrictor is not clear
- the expectation: the timing of distributivity and spreading errors should coincide
  - but this is not the case: Koster, Spenader, and Hendriks (2018)
  - spreading errors (Dutch children) disappear after 9 years
  - but in this age (until 11) children still don't acquire the correct interpretation of $\forall$

## Another explanation

- building on Dotlacil (2010): acquisition studies Pagliarini, Fiorin, and Dotlačil (n.d.), Koster, Spenader, and Hendriks (2018)
- conversational implicature explanation (after Horn (2001))
  - (9-a): non-ambiguously distributive
  - (9-b): principially ambiguous but (9-a) is logically stronger (asymetrically entails) and non-ambiguous
  - cooperative speaker knowing that (9-a) would be true would use it
  - → (9-a) is false and (9-b) has only the non-distributive interpretation

(9) a. Each boy drew two animals.
b. Two boys drew two animals.

- adult speakers speakers generally prefer non-distributive interperation for numerical NPs, definite NPs, ...:

(10) a. Each boy drew two animals.
b. Two boys drew two animals.
c. The boys drew two animals.

- first acquisition experiment: formal semantics and acquisition for Czech
- partially following the previous studies
- research question 1:

(11) What is the learning curve of UQs/bareNPs in Czech?

- we didn’t focus on the distributive vs. collective interpretation of UQs and definite NPs
- we scrutinized Czech UQs and bareNPs concerning their distributive and cumulative interpretation
- definiteness in Czech as a grammatical category is questionable at least: Šimík and Demian (2020)

## Example conditions

(12)

| | | | | | |
|---|---|---|---|---|---|
| Každé | dítě | si | koupilo | dva | bonbóny. |
| each | child | REFL | buy.3.SG.PAST | two | sweet.PL |

‘Each child bought two sweets.’

(13)

| | | | | |
|---|---|---|---|---|
| Děti | si | koupily | dva | bonbóny. |
| child.PL | REFL | buy.3.PL.PAST | two | sweet.PL |

‘Children bought two sweets.’

- link to the experiment (copy):

PCIbex

- the experimental platform for linguistics (University of Pennsylvania)

- research question 2:

(14) Can the acquisition level of Czech UQs predict the rejection of the distributive reading of bareNPs?

- positive answer can help to decide between the theories of distributivity acquisition errors

- 214 Czech children (age range: 5 - 11, median age: 7) passed the fillers of the experiment
    - on tablet (in kindergartens, schools)
    - pilot study: around 30 children, changes in design
        - shortened (no context), made more straightforward
        - pilot study data not used
- 47 Czech adults (median age: 24) also passed the fillers (online on PCIbex)
    - shared via link, on PC

## Design

- 2 x 2 design:
  - universal quantifier (UNIV) or bare NP (BARE) was tested against
  - distributive (DISTR) or cumulative (CUMUL) interpretation (pictures)

(15) Každé dítě si koupilo dva bonbóny.
each child REFL buy.3.SG.PAST two sweet.PL
'Each child bought two sweets.' UNIV

(16) Děti si koupily dva bonbóny.
child.PL REFL buy.3.PL.PAST two sweet.PL
'Children bought two sweets.' BARE

**Figure 4:** Test picture – distributive reading

**Figure 5:** Test picture – cumulative reading

- the learning curve for tested children: descriptively visualized in Fig. XX
  - first row of legend: the expression
  - the second row: the interpretation

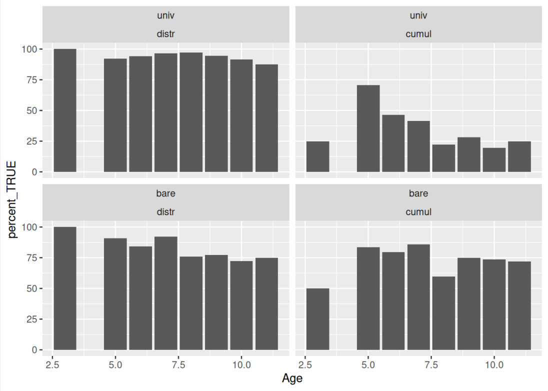



**Figure 6:** Children learning curve

- the adult control group descriptive statistics is in Figure XX:
    - x-axis: the expressions,
    - color: the interpretation

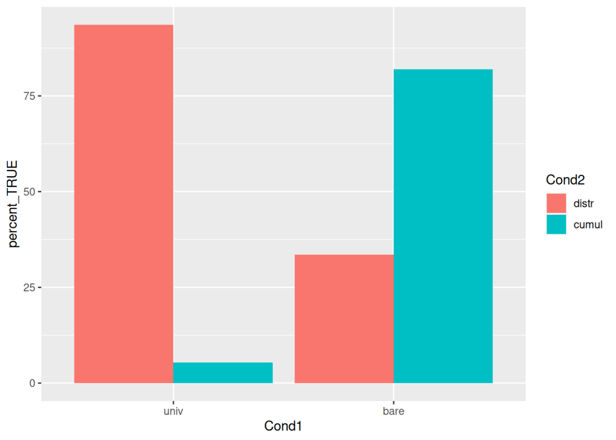



**Figure 7:** Adults control group

## Model

- Bayesian logistic hierarchical model with random effects and default priors was fit using the R package RSTANARM Goodrich et al. (2022):
- the dependent variable was the subject's answer:
  - checkmark: the image corresponds to the sentence vs.
  - cross: the image and the sentence don't correspond
- the independent variables were:
  - the condition (UNIVDISTR, UNIVCUMUL, BAREDISTR, BARECUMUL)
  - age of the subject (in case of children)
  - and their interaction
- the reference level was BARECUMUL

## Inferential statistics

A) adults:

- the distributive interpretation of UQs is acceptable to the same extent as the cumulative interpretation of bareNPs – reference level:
  - $\hat{\mu} = 0.12$, 95% Credibility Interval, CI = $[0.05, 0.18]$, 32.29% in ROPE
- distributive interpretation of bareNPs is much worse than the reference level
  - $\hat{\mu} = -0.48$, CI = $[-0.56, -0.41]$, 0% in ROPE

## Inferential statistics

A) adults:

- cumulative interpretation of UQs is also much worse than ref. level
  - $\hat{\mu} = -0.77$, CI = $[-0.84, -0.69]$, 0% in ROPE
- posterior samples graph for adults Bayes model on the next slide

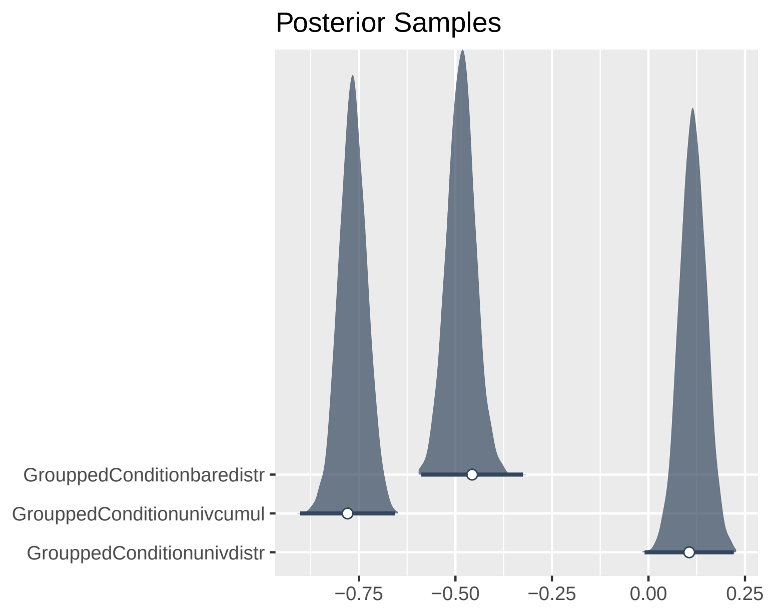



**Figure 8:** Bayesian model

```
Summary of Posterior Distribution

Parameter            | Median |          95% CI |     pd |          ROPE | % in ROPE |  Rhat |
-------------------------------------------------------------------------------------------------
(Intercept)          |   0.82 | [ 0.76,  0.87] |   100% | [-0.10, 0.10] |        0% | 0.999 | 364
Conditionbaredistr   |  -0.48 | [-0.56, -0.41] |   100% | [-0.10, 0.10] |        0% | 1.000 | 423
Conditionunivcumul   |  -0.77 | [-0.84, -0.69] |   100% | [-0.10, 0.10] |        0% | 1.000 | 417
Conditionunivdistr   |   0.12 | [ 0.05,  0.18] | 99.90% | [-0.10, 0.10] |    32.29% | 1.000 | 444
```

## Inferential statistics

B) children:

- the distributive interpretation of UQs is the same as the reference level
  - $\hat{\mu} = -0.13$, CI = $[-0.42, 0.13]$, 40.29% in ROPE
- and this remains stable during the acquisition
  - non-credible interaction effect UnivDistr by AGE: $\hat{\mu} = 0.15$, CI = $[0.01, 0.30]$, 22.82% in ROPE

## Inferential statistics

B) children

- unlike adults, children accept the distributive interpretation of bareNPs to the same extent as the reference level
  - $\hat{\mu} = 0.21$, CI = $[-0.07, 0.47]$, 21.21% in ROPE
- and at least in the sample the development towards adults state is non-credible
  - the interaction between BAREDISTR and AGE: $\hat{\mu} = -0.07$, CI = $[-0.21, 0.07]$, 64.03% in ROPE

## Inferential statistics

B) children

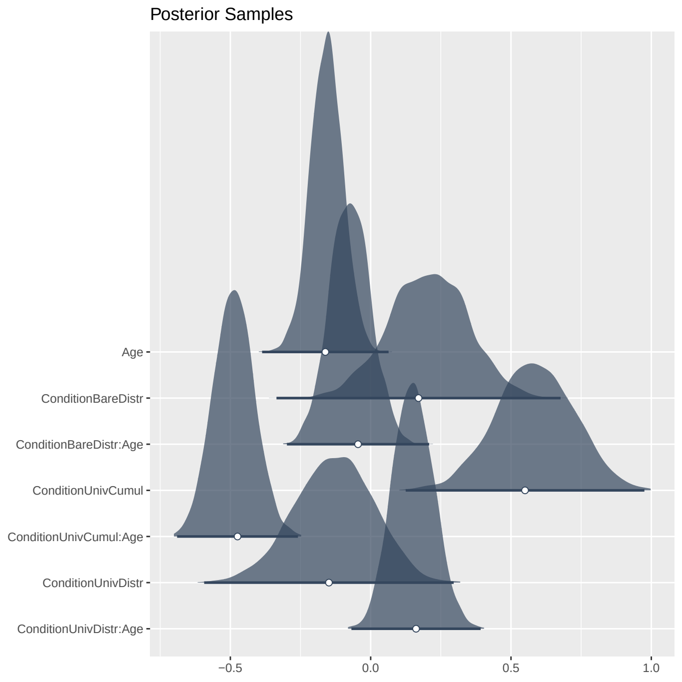
- cumulative interpretation of UQs is as a main effect indistinguishable from the reference level
  - $\hat{\mu} = 0.58$, CI = $[0.31, 0.85]$, 0% in ROPE
- but there is a clear development towards adult grammar
  - the interaction between UNIVCUMUL and AGE is a strong negative effect: $\hat{\mu} = -0.49$, CI = $[-0.63, -0.34]$, 0% in ROPE

## Inferential statistics

B) children

- the cumulative interpretation of bareNPs is strongly acceptable
  - ref. level – the intercept: $\hat{\mu}$ = 1.08, CI = [0.86, 1.30]
- and doesn't change during acquisition
  - the main effect of AGE is non-credible: $\hat{\mu}$ = −0.16, CI = [−0.27, −0.04], 14.42% in ROPE
- posterior samples graph for adults Bayes model on the next slide

report


Posterior Samples
Age
ConditionBareDistr
ConditionBareDistr:Age
ConditionUnivCumul
ConditionUnivCumul:Age
ConditionUnivDistr
ConditionUnivDistr:Age
−0.5
0.0
0.5
1.0

```
describe_posterior(model_bayes2)

Summary of Posterior Distribution

Parameter                | Median |         95% CI |     pd |          ROPE | % in ROPE |  Rhat |
-------------------------------------------------------------------------------------------------
(Intercept)              |   1.08 | [ 0.86,  1.30] |   100% | [-0.10, 0.10] |        0% | 1.001 |
ConditionBareDistr       |   0.21 | [-0.07,  0.47] | 92.55% | [-0.10, 0.10] |    21.21% | 1.000 |
ConditionUnivCumul       |   0.58 | [ 0.31,  0.85] |   100% | [-0.10, 0.10] |        0% | 1.001 |
ConditionUnivDistr       |  -0.13 | [-0.42,  0.13] | 82.15% | [-0.10, 0.10] |    40.29% | 1.001 |
Age                      |  -0.16 | [-0.27, -0.04] | 99.72% | [-0.10, 0.10] |    14.42% | 1.001 |
ConditionBareDistr:Age   |  -0.07 | [-0.21,  0.07] | 84.82% | [-0.10, 0.10] |    64.03% | 1.001 |
ConditionUnivCumul:Age   |  -0.49 | [-0.63, -0.34] |   100% | [-0.10, 0.10] |        0% | 1.001 |
ConditionUnivDistr:Age   |   0.15 | [ 0.01,  0.30] | 98.65% | [-0.10, 0.10] |    22.82% | 1.001 |
```

# Theoretical consequences

- Czech children acquire (age: 5 to 11) the correct interpretation of the UQs
- they start with accepting both its distributive and cumulative interpretation
- but end with the adult-like rejection of the second interpretation (the answer to question 1):

(17) What is the learning curve of UQs/bareNPs in Czech?

- see the top right right corner in Figure XX and the Bayesian model in XX

- the acqusition of the UQ’s correct interpretation must precede adults’ conversational implicature stage
  - where the distributive interpretation of numerical NPs is (mostly) rejected (suggests a positive answer to question 2):

(18) Can the acquisition level of Czech UQs predict the rejection of the distributive reading of bareNPs?

- we didn’t test for spreading errors but the spreading hypothesis doesn’t predict any timing between UQ and bare NPs acquisition sequence
  - and also has wrong cross-linguistic predictions (see Koster, Spenader, and Hendriks (2018))

- decreasing trend of distributive interpretation for bare NPs (see the bottom left corner in Figure 1), but in our sample, the inferential effect was not credible
- the distributive interpretation for bare NPs is rejected after 11 years, reaching (at the end) the adult state in Figure XX
- this follows the conversational implicature hypothesis, which predicts that the rejection of the distributive interpretation for bare NPs must follow after the correct acquisition of the universal quantifier meaning

Thanks!

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